

# T1 Run Report: 2026-10-03-recorded

What ran, what each experiment observed, and every check, generated from the recorded run directory.

T1

**T1 · Production AI Engineering**

8 min read · Published 2026-10-03 · Run report

## About this report

**RECORDED**

generated by tools/build\_results.py from agent\_identity\_poc/runs/2026-10-03-recorded/

This is the forensic record behind both editions. It is generated from the run directory: tables are rendered from the experiment files and headline numbers are substituted from `facts.json`.

Nothing here is typed. For each claim traced to its evidence, with a status, see the Evidence Check; for what is real code and what is simulated, see Real vs simulated; to step through any scenario, see the Lab Console.

**41 of 41 declared checks passed** (30 experiment checks and 11 global assertions; 0 failed). The audit chain in `audit.jsonl` verifies link by link.

## What ran

The same rollback, under three identity models: **A** shared service account, **B** user token (impersonation), **C** delegation chain. Held constant: the incident (payment-service, 14% errors, the Datadog event dd-evt-771204); the proposed rollback (payment-service, production, v4.18.0 → v4.17.2); the logical agents and their deterministic plans; the policy fixture (config/policies.yaml); the approval fixture (ic.dev approves the exact call); the tool simulators and their logs (aid/tools.py); the run configuration; the nine attribution questions; the scenarios. The only variable: identity propagation and credential architecture.

# Environment, declared criteria and integrity

| Item                                             | Value                                                                              |
|--------------------------------------------------|------------------------------------------------------------------------------------|
| Run                                              | 2026-10-03-recorded                                                                |
| Recorded at                                      | 2026-10-03T18:25:54Z                                                               |
| Criteria frozen at                               | 2026-10-03T18:25:33Z ( proof/FREEZE.json )                                         |
| Source commit                                    | unknown (not a git repository)                                                     |
| Python                                           | 3.12.13                                                                            |
| Randomness                                       | none: simulated clock, directory, attestation and tools; deterministic agent plans |
| Agents                                           | deterministic plans (aid/agents.py); no model                                      |
| Tests (pytest)                                   | 34                                                                                 |
| Checks                                           | 41 (30 experiment, 11 global)                                                      |
| Scenarios recorded                               | 31                                                                                 |
| Policy fixture                                   | sha256 87fc40ea63965faf...                                                         |
| Scenario fixture                                 | sha256 256c20009786395f...                                                         |
| Identity fixture principals.yaml                 | sha256 47af27c0a6a641c8...                                                         |
| Identity fixture tool_identities.yaml            | sha256 1804394fb40e1b0b...                                                         |
| Identity fixture shared_sa.yaml                  | sha256 1405668e90d30951...                                                         |
| Frozen config/capabilities.yaml                  | sha256 d846724b16c40702...                                                         |
| Frozen config/policies.yaml                      | sha256 87fc40ea63965faf...                                                         |
| Frozen config/principals.yaml                    | sha256 47af27c0a6a641c8...                                                         |
| Frozen config/shared_sa.yaml                     | sha256 1405668e90d30951...                                                         |
| Frozen config/tool_identities.yaml               | sha256 1804394fb40e1b0b...                                                         |
| Frozen criteria (checks_for + global_assertions) | sha256 139f529874bd8fe4...                                                         |
| Frozen proof/preregistration.toml                | sha256 1facdf0cb93023c6...                                                         |

The 29 original checks pre-date this run; their results were already known from run 2026-09-29-recorded. They were kept unchanged, one I4 check and the eleven global assertions G01-G11 were

added, and all 41 were declared in `proof/preregistration.toml` and frozen before this run. `aid` experiments refuses to run if a frozen file changed, or if the checks it evaluates differ from the declared list. Changes after the freeze ( `proof/DEVIATIONS.md` ):

| # | When (UTC)           | What changed                                                                                                          | Why                                                                                                                                                                                 | Effect on a check       |
|---|----------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| 1 | 2026-10-03T18:25:54Z | <code>aid/experiments.py</code><br><code>collected_tests()</code> : read pytest's per-file <code>file: N</code> lines | Before the first run after the freeze, the manifest's test count came out as -1: <code>pytest --collect-only -q</code> prints per-file counts, not a total. Manifest metadata only. | None: no check reads it |

The run is deterministic. Re-running the experiments into a fresh copy of the POC reproduces this directory byte for byte, except `recorded.json` , which says when the run was recorded ( `make verify` ).

## Scorecard

| Experiment                                   | Checks | Passed | Kinds                         |
|----------------------------------------------|--------|--------|-------------------------------|
| I1 · Attribution completeness                | 7      | 7      | control, invariant            |
| I2 · Confused deputy                         | 4      | 4      | control, invariant            |
| I3 · Revocation granularity and blast radius | 7      | 7      | control, invariant, qualified |
| I4 · Replay and workload binding             | 6      | 6      | control, invariant, qualified |
| I5 · Impersonation provenance loss           | 3      | 3      | control, invariant            |
| I6 · Privilege build-up                      | 1      | 1      | invariant                     |
| I7 · Long approval pause                     | 2      | 2      | control, invariant            |
| GA · Global assertions                       | 11     | 11     | invariant                     |

Kinds: **invariant**, a property the delegation chain must hold; **control**, a weak model predicted to lose the property (passing means it did); **qualified**, a property that holds only within a stated bound.

## I1 · Attribution: the nine questions, from each record

The same rollback under each identity model. The nine questions are declared in `proof/preregistration.toml` (A1–A9). A question counts as answered only if the record states the true value.

| Question                              | Shared account: platform | User token: platform | Delegation chain: platform | Shared account: tool log | User token: tool log | Delegation chain: tool log |
|---------------------------------------|--------------------------|----------------------|----------------------------|--------------------------|----------------------|----------------------------|
| A1 Which agent acted?                 | X                        | X                    | ✓                          | X                        | X                    | X                          |
| A2 Who invoked it?                    | X                        | X                    | ✓                          | X                        | X                    | X                          |
| A3 On whose behalf?                   | X                        | ✓                    | ✓                          | X                        | ✓                    | X                          |
| A4 With what authority (scopes)?      | X                        | X                    | ✓                          | X                        | X                    | X                          |
| A5 On which runtime workload?         | X                        | X                    | ✓                          | X                        | X                    | X                          |
| A6 With which tool credential?        | ✓                        | ✓                    | ✓                          | ✓                        | ✓                    | ✓                          |
| A7 What action, with which arguments? | ✓                        | ✓                    | ✓                          | ✓                        | ✓                    | ✓                          |
| A8 Who approved it?                   | ✓                        | ✓                    | ✓                          | X                        | X                    | X                          |
| A9 Can each part be revoked?          | X                        | X                    | ✓                          | X                        | X                    | X                          |
| <b>Answered</b>                       | <b>3/9</b>               | <b>4/9</b>           | <b>9/9</b>                 | <b>2/9</b>               | <b>3/9</b>           | <b>2/9</b>                 |

| Identity model   | Kubernetes saw                                     | Agents acting | Attributable from the platform record | Distinct Kubernetes principals |
|------------------|----------------------------------------------------|---------------|---------------------------------------|--------------------------------|
| Shared account   | system:serviceaccount:platform:ai-automation       | 3             | 0                                     | 1                              |
| User token       | maya@company.com                                   | 3             | 0                                     | 2                              |
| Delegation chain | system:serviceaccount:payments:incident-remediator | 3             | 3                                     | 2                              |

Tamper test: the approver in row 8 of 8 was edited after the fact; the chain verified before the edit (✓) and failed at row 8 after it.

| Id    | Check                                                          | Kind      | Arm | Result |
|-------|----------------------------------------------------------------|-----------|-----|--------|
| I1-01 | the rollback executed once in every mode                       | invariant | ABC | pass   |
| I1-02 | chain: the platform record answers all nine questions          | invariant | C   | pass   |
| I1-03 | every mode: the tool log answers at most three                 | invariant | ABC | pass   |
| I1-04 | shared account: Kubernetes sees one principal for three agents | control   | A   | pass   |
| I1-05 | shared account: no agent attributable from the platform record | control   | A   | pass   |
| I1-06 | chain: all three agents attributable from the platform record  | invariant | C   | pass   |
| I1-07 | audit chain intact; an edited approver breaks it at that row   | invariant | C   | pass   |

## I2 · The confused deputy

`agent.release-guard` (read only) asks `agent.incident-intel` to roll back production.

| Identity model   | First decision   | Rule                    | Approver saw            | Originator visible | Rollbacks |
|------------------|------------------|-------------------------|-------------------------|--------------------|-----------|
| Shared account   | REQUIRE_APPROVAL | P2-high-risk-production | agent.incident-intel    | ✗                  | 1         |
| User token       | REQUIRE_APPROVAL | P2-high-risk-production | agent.incident-intel    | ✗                  | 1         |
| Delegation chain | DENY             | P1-missing-scope        | (no approval requested) | ✓                  | 0         |

Delegation chain: caller scopes `deploy:read, itsm:read, release:check, telemetry:read`; callee scopes after narrowing `deploy:read, itsm:read, telemetry:read`; actor chain `agent.incident-intel@1.3.0` → `agent.release-guard@0.9.2`; an undeclared agent → agent edge refused: ✓.

| Id    | Check                                                                 | Kind      | Arm | Result |
|-------|-----------------------------------------------------------------------|-----------|-----|--------|
| I2-01 | chain: the deputy request is denied before any approval               | invariant | C   | pass   |
| I2-02 | chain: the callee's scopes narrowed                                   | invariant | C   | pass   |
| I2-03 | chain: an undeclared agent -> agent edge is refused                   | invariant | C   | pass   |
| I2-04 | shared account: the deputy rollback executes and hides its originator | control   | A   | pass   |

## I3 · Revocation: one lever, five running executions

Executions: **E1** event -> incident-intel; **E2** maya (web) -> incident-intel; **E3** ci -> release-guard (release-runtime); **E4** workflow -> incident-intel; **E5** maya (web) -> incident-intel -> remediation. Each lever was pulled at minute 5 of a 30-minute window; execution tokens live 15 minutes.

| Lever                   | Stopped            | Kept running   | Minutes to effect            | New start afterwards                                  |
|-------------------------|--------------------|----------------|------------------------------|-------------------------------------------------------|
| agent                   | E1, E2, E4, E5     | E3             | E1 0, E2 0, E4 0, E5 0       | refused: agent agent.incident-intel@1.3.0 is disabled |
| delegation              | E2, E5             | E1, E3, E4     | E2 10, E5 10                 | accepted                                              |
| invoker credential      | E1                 | E2, E3, E4, E5 | E1 10                        | refused: unknown or revoked credential                |
| tool identity           | E5                 | E1, E2, E3, E4 | E5 0                         | accepted                                              |
| workload                | E1, E2, E4, E5     | E3             | E1 0, E2 0, E4 0, E5 0       | accepted                                              |
| shared account rotation | E1, E2, E3, E4, E5 | none           | E1 0, E2 0, E3 0, E4 0, E5 0 | accepted                                              |

| Id    | Check                                                              | Kind      | Arm | Result |
|-------|--------------------------------------------------------------------|-----------|-----|--------|
| I3-01 | control: nothing stops without a revocation                        | invariant | C   | pass   |
| I3-02 | delegation: only Maya's executions stop, within one token lifetime | qualified | C   | pass   |
| I3-03 | invoker credential: new executions from that head refused          | invariant | C   | pass   |
| I3-04 | agent: every execution running that agent stops at its next call   | invariant | C   | pass   |
| I3-05 | workload: executions on the other runtime keep running             | invariant | C   | pass   |
| I3-06 | tool identity: one capability of one execution stops               | invariant | C   | pass   |
| I3-07 | shared account rotation stops all five executions                  | control   | A   | pass   |

# I4 · Replay and theft

| Attempt                                                                                                       | Outcome  | Reason                                                                                                                |
|---------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------|
| Legitimate call                                                                                               | EXECUTED | -                                                                                                                     |
| Execution token from another workload                                                                         | REJECTED | token bound to spiffe://prod.company.internal/agent-runtime, presented by spiffe://prod.company.internal/batch-runner |
| Approved rollback's token and approval from another workload (rollbacks: 0; then the real workload: EXECUTED) | REJECTED | token bound to spiffe://prod.company.internal/agent-runtime, presented by spiffe://prod.company.internal/batch-runner |
| Expired execution token                                                                                       | REJECTED | token expired                                                                                                         |
| Tool credential at the wrong audience                                                                         | 401      | credential audience is kubernetes, not jira                                                                           |
| Stolen tool credential, within its 10 minutes                                                                 | 200      | ok                                                                                                                    |
| Stolen tool credential, after expiry                                                                          | 401      | credential expired or revoked                                                                                         |
| Shared secret, from anywhere, a day later                                                                     | 200      | ok                                                                                                                    |

| Id    | Check                                                                           | Kind      | Arm | Result |
|-------|---------------------------------------------------------------------------------|-----------|-----|--------|
| I4-01 | execution token replayed from another workload is rejected                      | invariant | C   | pass   |
| I4-02 | privileged write replayed from another workload is rejected with no side effect | invariant | C   | pass   |
| I4-03 | expired execution token is rejected                                             | invariant | C   | pass   |
| I4-04 | tool credential rejected by a system it was not minted for                      | invariant | C   | pass   |
| I4-05 | stolen tool credential dies with its ten minutes                                | qualified | C   | pass   |
| I4-06 | shared secret still works from anywhere a day later                             | control   | A   | pass   |

# I5 · Impersonation vs delegation

|                                                                  | User token (impersonation) | Delegation chain |
|------------------------------------------------------------------|----------------------------|------------------|
| Agent recorded                                                   | ✗                          | ✓                |
| Actor chain recorded                                             | ✗                          | ✓                |
| Agent's action indistinguishable from Maya's own in the tool log | ✓                          | ✗                |
| Platform record fields                                           | 9                          | 26               |
| Tool permissions behind the call                                 | 5                          | 3                |

Identity fields lost by impersonation (13): `act_chain` , `agent` , `event_id` , `event_rule` , `event_source` , `grant_id` , `invoker` , `invoker_credential` , `revocation` , `scopes` , `token_id` , `tool_identity` , `workload` .

| Id    | Check                                                                  | Kind      | Arm | Result |
|-------|------------------------------------------------------------------------|-----------|-----|--------|
| I5-01 | impersonation: agent action identical to Maya's own in Kubernetes' log | control   | B   | pass   |
| I5-02 | delegation: agent action distinguishable from Maya's own               | invariant | C   | pass   |
| I5-03 | impersonation loses the agent and the actor chain                      | invariant | BC  | pass   |

## I6 · Privilege accumulation (derived from configuration)

DERIVED

 computed from `config/shared_sa.yaml` and `config/tool_identities.yaml`, not measured at run time

| Measure                                   | Value |
|-------------------------------------------|-------|
| Permissions held by the shared account    | 20    |
| Grants on the shared account              | 11    |
| Grants that exist only for retired agents | 5     |
| Permissions only retired agents needed    | 11    |
| Permissions the platform needs today      | 6     |
| Excess permissions                        | 14    |
| Tool identities in the chain model        | 5     |
| Most permissions on any one tool identity | 3     |

| Id    | Check                                            | Kind      | Arm | Result |
|-------|--------------------------------------------------|-----------|-----|--------|
| I6-01 | shared account holds more than any tool identity | invariant | AC  | pass   |

# I7 · The pause: revoked while waiting for approval

| Variant               | Approval requested | Approved | Final    | Rollbacks | Reason                                                |
|-----------------------|--------------------|----------|----------|-----------|-------------------------------------------------------|
| Shared account        | REQUIRE_APPROVAL   | ✓        | EXECUTED | 1         | ok                                                    |
| Chain, token extended | REQUIRE_APPROVAL   | ✓        | EXECUTED | 1         | ok                                                    |
| Chain, re-exchanged   | REQUIRE_APPROVAL   | ✓        | REJECTED | 0         | re-exchange refused: delegation from sre.maya revoked |

The delegation was revoked at minute 10 of a 37-minute approval wait.

| Id    | Check                                                         | Kind      | Arm | Result |
|-------|---------------------------------------------------------------|-----------|-----|--------|
| I7-01 | re-exchange after the pause: revoked delegation stays revoked | invariant | C   | pass   |
| I7-02 | extending instead of re-exchanging lets the rollback through  | control   | C   | pass   |

## Global assertions (G01–G11)

| Id  | Check                                                                                           | Kind      | Arm | Result |
|-----|-------------------------------------------------------------------------------------------------|-----------|-----|--------|
| G01 | delegation never widens authority (the callee's scopes are within the caller's after every hop) | invariant | C   | pass   |
| G02 | a delegated write whose originator lacks the authority is denied, not silently executed         | invariant | C   | pass   |
| G03 | a credential replayed from the wrong workload causes no privileged side effect                  | invariant | C   | pass   |
| G04 | a revoked identity cannot mint new execution authority                                          | invariant | C   | pass   |
| G05 | unrelated executions survive every targeted revocation                                          | invariant | C   | pass   |
| G06 | rotating the shared account stops more executions than any targeted lever                       | invariant | AC  | pass   |
| G07 | agent identity and runtime identity stay distinct in the record                                 | invariant | C   | pass   |
| G08 | impersonation does not pass for delegation provenance                                           | invariant | C   | pass   |
| G09 | no tool log is complete upstream provenance (every tool log answers fewer than nine questions)  | invariant | ABC | pass   |
| G10 | the platform record reconstructs the whole identity chain (nine of nine)                        | invariant | C   | pass   |
| G11 | tampering with the platform audit is detected at the edited row                                 | invariant | C   | pass   |

## Supported, qualified, contradicted

Every declared check passed, so nothing contradicted an expectation (0 failed). The checks of kind *qualified* hold only within a bound: I3-02 (delegation: only Maya's executions stop, within one token lifetime), I4-05 (stolen tool credential dies with its ten minutes). Each claim's status, and the qualified findings that are not a check (a stolen tool credential works at its one audience until it expires; the tool log holds only the edge; the hash chain is not an anchored ledger; I6 is derived from configuration), are in the Evidence Check.

## Reproduce

```
cd agent_identity_poc && uv sync --group dev
uv run pytest                                # 34 tests, including test_experiments.py
(I1-I7, G01-G11)
uv run aid experiments --run-id 2026-10-03-recorded # refuses if a frozen file changed
cd .. && make verify                          # rerun into a fresh copy; byte for byte
```

# Files behind this report

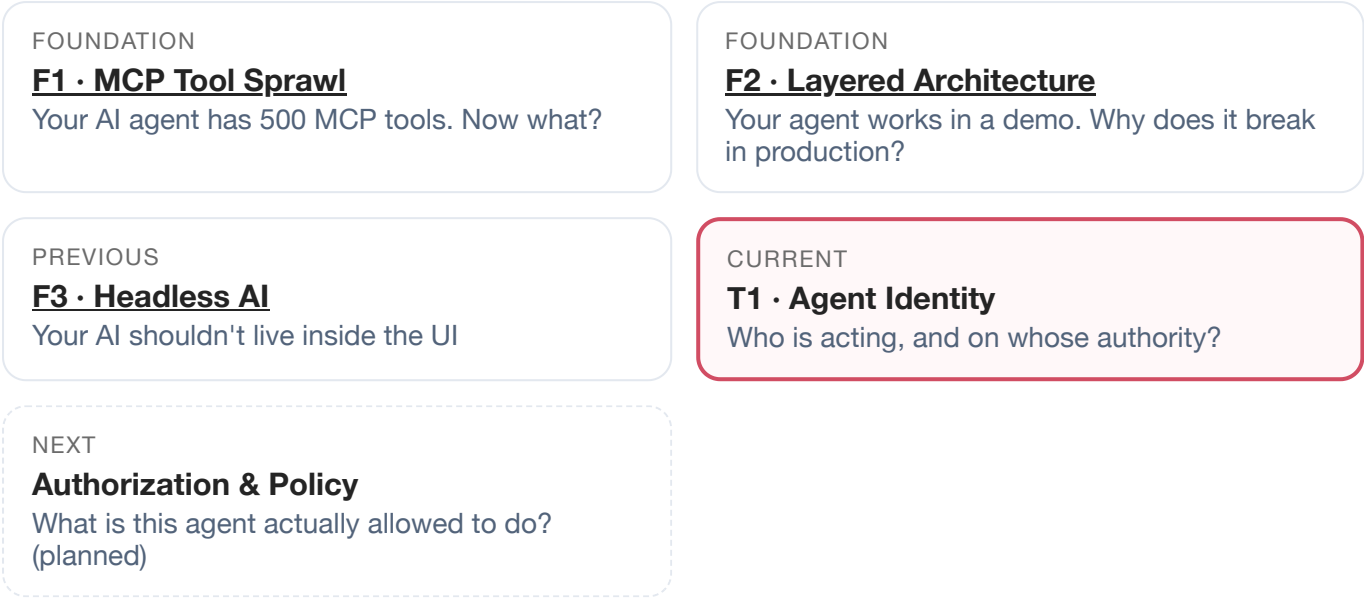
| File           | sha256              |
|----------------|---------------------|
| I1.json        | 9cd981dd5868380a... |
| I2.json        | bcc7e2a95e1f91b7... |
| I3.json        | b0b2ae423c99f4bc... |
| I4.json        | f906000b2a8bdce8... |
| I5.json        | fb3466ab1d3dada1... |
| I6.json        | 50ca3d94799731dc... |
| I7.json        | bd17557bdd7892fa... |
| audit.jsonl    | f8979c4eaf2818f4... |
| checks.json    | 5d8da5d254b4229d... |
| facts.json     | 9faf1187cb9e2571... |
| manifest.json  | 60ca6f4b69236b49... |
| recorded.json  | b9e2a9e2b8969755... |
| story.json     | 6db90a07a47a93fe... |
| summary.md     | 5bf27b191a1b5a22... |
| tool-logs.json | 6ef678c13c0721a9... |

Evidence

Run report

Agent identity

PRODUCTION AI ENGINEERING · THE SERIES SO FAR



[The full learning map \(Start Here\)](#)

# T1 · Production AI Engineering

Production AI Engineering · T1 · Evidence. Every measured number here is substituted at build time from `agent_identity_poc/runs/2026-10-03-recorded/facts.json`; the enterprise systems are simulated and the run is deterministic. Figures carry their provenance.

[Medium edition](#)   [Technical deep dive](#)   [Evidence Check](#)   [POC README](#)   [Run summary](#)  
[QA report](#)